Franklin University - ECON 251 (Intermediate Microeconomics)
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Thursday, July 16th
Assignment Title - "Homework Module 9"

Problem 1

1. Production function Y(l) = l^α, α > 0

(a) Marginal product

MP = dY/dl = α·l^(α−1). The sign of the exponent (α − 1) determines how marginal product behaves as labor rises: for 0 < α < 1 marginal product is decreasing (diminishing returns); for α = 1 it is constant (MP = 1, since Y = l is linear); for α > 1 it is increasing (increasing returns to the labor input).

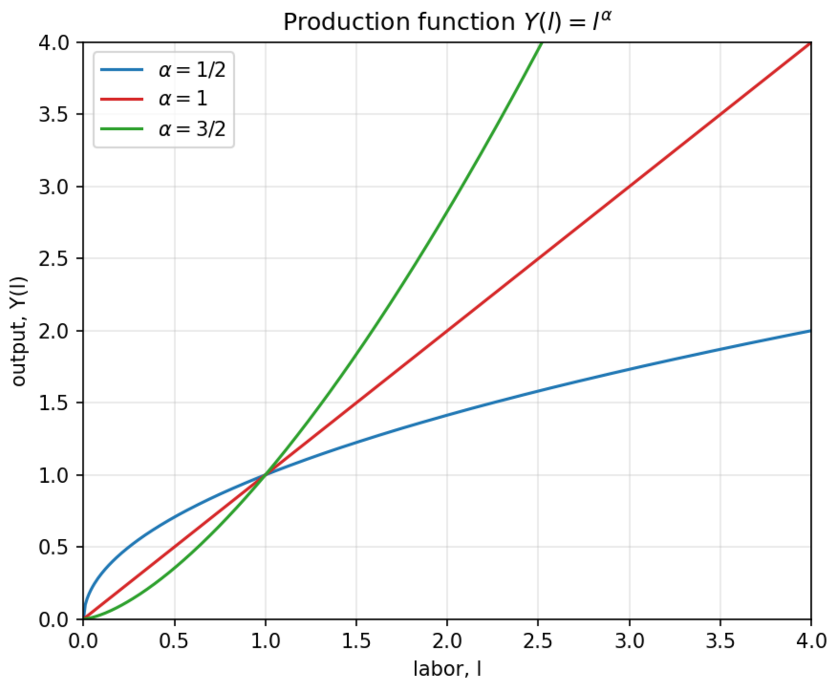

Figure 1. Y(l) = l^α for α = 1/2 (concave, diminishing MP), α = 1 (linear), α = 3/2 (convex, increasing MP).

(b) Cost minimization

With labor the only input at wage w > 0, cost is C = w·l. Inverting the production function, l = Y^(1/α), so:

C(Y) = w·Y^(1/α)

MC(Y) = dC/dY = (w/α)·Y^(1/α − 1)

AC(Y) = C(Y)/Y = w·Y^(1/α − 1)

(c) Comparing AC and MC

Both AC and MC share the exponent (1/α − 1), so they move together: for α < 1, 1/α − 1 > 0 and both AC and MC are increasing in y; for α = 1, the exponent is 0 and both are constant (AC = MC = w); for α > 1, the exponent is negative and both are decreasing in y. This is intuitive: increasing marginal product (α > 1) translates into falling marginal and average cost as output rises, and vice versa for α < 1.

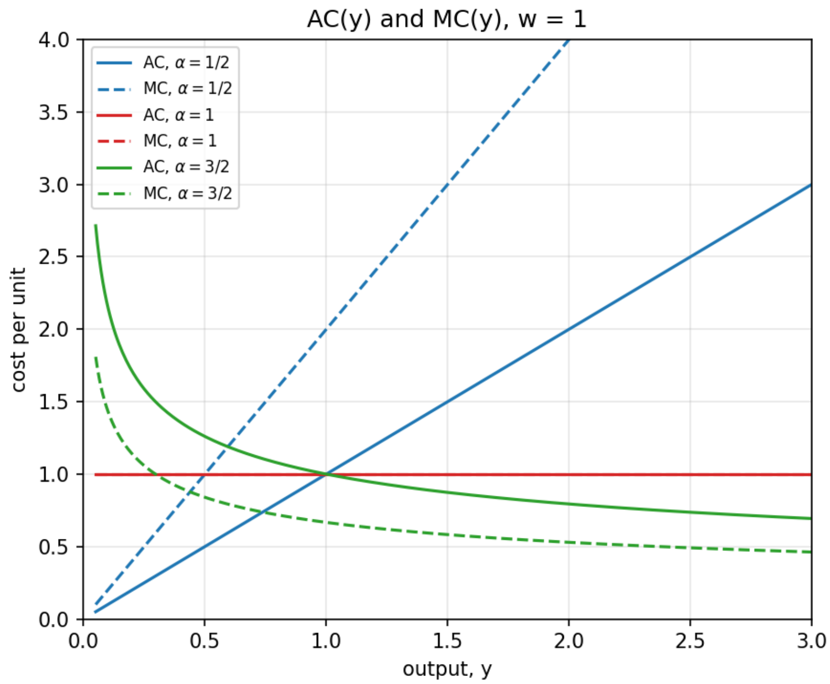

Figure 2. AC(y) (solid) and MC(y) (dashed) for α = 1/2, 1, 3/2, with w = 1.

2. Production function Y(l) = (l − γ)^β, β ∈{1/2, 1}, γ > 0

(a) Cost functions

Here γ is a fixed amount of labor needed before any output is produced (a technologically fixed input requirement). Inverting: l = γ + Y^(1/β), so total cost is:

$C(Y) = w\cdot\gamma + w\cdot Y^{(1/\beta)}$ (w·γ is a fixed cost)

$MC(Y) = (w/\beta)\cdot Y^{(1/\beta - 1)}$

$AC(Y) = w\cdot\gamma/Y + w\cdot Y^{(1/\beta - 1)}$

(b) The two special cases

For β = 1/2: 1/β = 2, so $C(y) = w\cdot\gamma + w\cdot y^2$, MC(y) = 2w·y (increasing, linear), and AC(y) = w·γ/y + w·y — the classic U-shape (falling while the spread-out fixed cost dominates, then rising as the w·y term dominates). For β = 1: 1/β = 1, so C(y) = w·γ + w·y, MC(y) = w (constant), and AC(y) = w·γ/y + w falls monotonically toward w as y → ∞ and never turns upward.

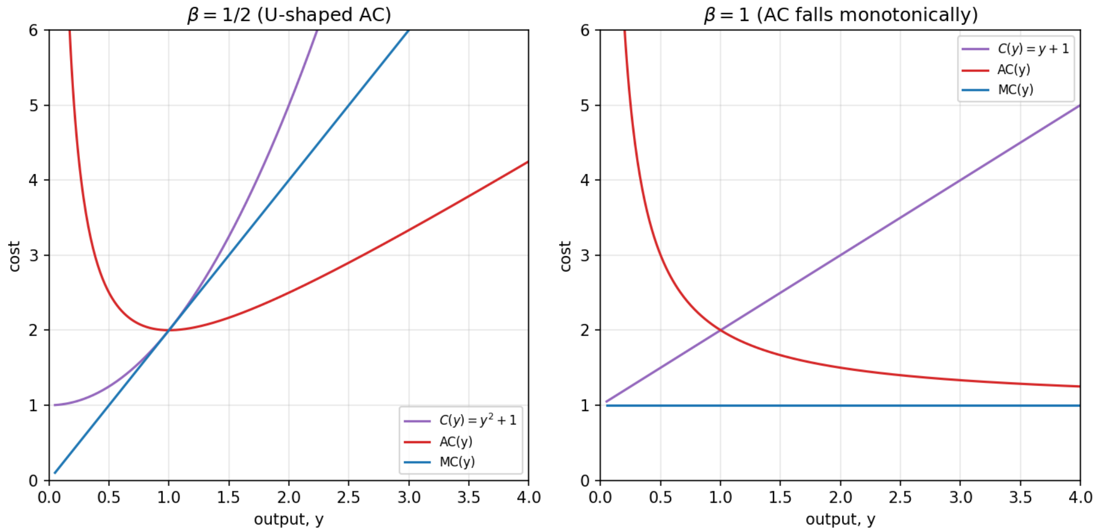

Figure 3. Cost, AC, and MC with w = γ = 1: β = 1/2 gives a U-shaped AC; β = 1 gives a monotonically falling AC with constant MC.

3. Which production function is incompatible with a perfectly competitive market?

A competitive market of many small price-taking firms requires each firm to have a well-defined efficient scale — i.e., an average cost curve that eventually rises, so firms don't have an unbounded incentive to keep growing. Judged against that requirement:

$Y = l^\alpha$: only $\alpha < 1$ (diminishing returns) gives the required eventually-rising cost structure. $\alpha \geq 1$ (constant or increasing returns to scale) is incompatible — cost per unit never rises, so bigger is always at least as cheap, pushing the market toward a single dominant firm (a natural monopoly), not perfect competition.

$Y = (l - \gamma)^\beta$: only $\beta = 1/2$ produces a U-shaped AC with an interior minimum, compatible with competition. $\beta = 1$ is incompatible for the same reason as above — AC falls forever, so there is no finite efficient scale and the technology favors ever-larger firms.

Problem 2

1. Maximizer vs. satisficer

A maximizer searches exhaustively for the objectively best option, which raises search costs, the opportunity cost of time, and post-decision regret from comparing the choice to every foregone alternative. A satisficer stops as soon as an option clears a “good enough” threshold. The satisficer can end up happier not because her holiday is objectively better, but because her expectations are calibrated to “good enough” rather than “the best,” so there is a smaller gap between expectation and experience, less counterfactual comparison, and far lower search/decision costs. This matches Herbert Simon's bounded-rationality account of satisficing and Barry Schwartz's empirical finding that maximizers report lower subjective well-being than satisficers even when they secure objectively better outcomes.

2. Tesla Model 3 (WirtschaftsWoche, May 2018)

Reported figures: price range \$35,000–\$78,000, material/delivery cost ≈ \$18,000, additional production cost ≈ \$10,000 per car, so total variable cost per car ≈ \$28,000. Even at the low end of the price range, the contribution margin is \$35,000 − \$28,000 = \$7,000 per car, and it rises to roughly \$50,000 at the top end — so on a per-unit basis, the numbers pointed to Tesla being able to cover variable costs and generate a positive contribution margin at every price point in that range, consistent with the article's claim that the Model 3 “may become profitable.”

That said, this is a variable-cost calculation only: it says nothing about whether revenue also covers fixed costs (factory capex, R&D amortization, overhead) or the implicit/opportunity cost of invested capital — the ingredients of economic profit rather than just contribution margin. A positive per-car margin is necessary but not sufficient for the firm as a whole to be economically profitable; it was a reasonable basis for cautious optimism, not proof of overall profitability.

Problem 3

Munger's claim — that intelligent decisions weigh opportunity cost, i.e., the value of the best foregone alternative — is the textbook definition of economic profit and rational choice (see Problem 4 below). Schwartz's quote points to the practical limit on that ideal: identifying and evaluating “unlimited possibilities” is itself costly in time, attention, and cognitive effort, and beyond some point, the cost of choosing well can exceed the benefit of the marginally better choice.

In my view, not all decisions should be made by exhaustively computing opportunity cost in the literal sense. Doing so is itself an activity with an opportunity cost (the time spent comparing options is time not spent doing something else), so for low-stakes, frequent, or reversible decisions, a satisficing heuristic — a rule of thumb that is good enough — is often the more rational choice in a bounded-rationality sense. Reserving careful opportunity-cost analysis for decisions that are high-stakes, rare, or hard to reverse (e.g., a career change or a major purchase) captures most of the benefit of Munger's principle without paying the full cost Schwartz describes.

Problem 4

Reading Fig. 13.1: MC and AC cross at their common minimum near $y = 10$, $p \approx 20$, and AVC appears to run linearly through the origin. This is exactly the shape derived in Problem 1.2(b) for $\theta = 1/2$. Taking $w = 1$ and $\gamma = 100$ reproduces the figure: $AVC(y) = y$, $AC(y) = y + 100/y$, $MC(y) = 2y$ ($AC = MC = 20$ at $y = 10$, matching the figure). The numeric values below follow from that reading of the graph; the method is what matters if the exact figures were read slightly differently.

1. Producer surplus at $\bar{p} = 10$

The firm's short-run supply is its MC curve (above shutdown): $p = MC(y) = 2y \Rightarrow y^* = p/2$. At $\bar{p} = 10$, $y^* = 5$, and since $\bar{p} = 10 > AVC(5) = 5$, the firm produces rather than shutting down. Producer surplus is the area between price and MC from 0 to y^*:

$$PS = \int_0^5 (10 - 2y)\, dy = [10y - y^2]_0^5 = 50 - 25 = 25$$

This is the shaded triangle in Figure 4 below (between the $\bar{p} = 10$ line and the MC curve, from $y = 0$ to $y = 5$).

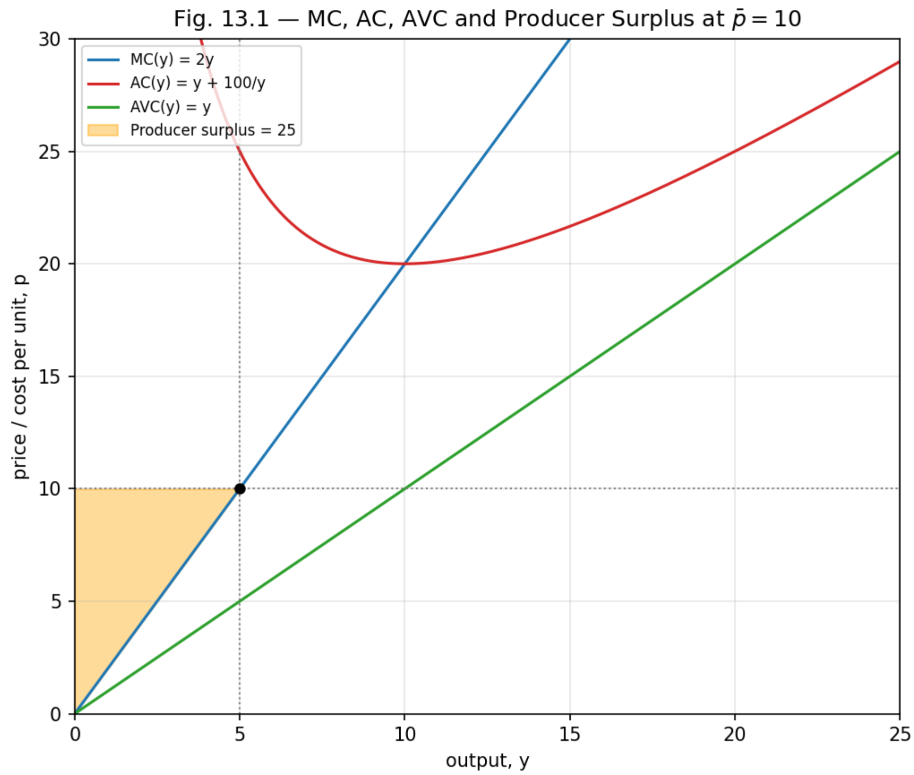

Figure 4. MC, AC, AVC and the producer-surplus triangle at $\bar{p} = 10$ ($y^* = 5$, PS = 25).

2. Average fixed cost AFC(y)

Since AFC(y) = AC(y) − AVC(y) = 100/y, it is a downward-sloping rectangular hyperbola: very large as y → 0 (the fixed cost is spread over almost no output) and approaching 0 as y grows (the fixed cost gets spread thin).

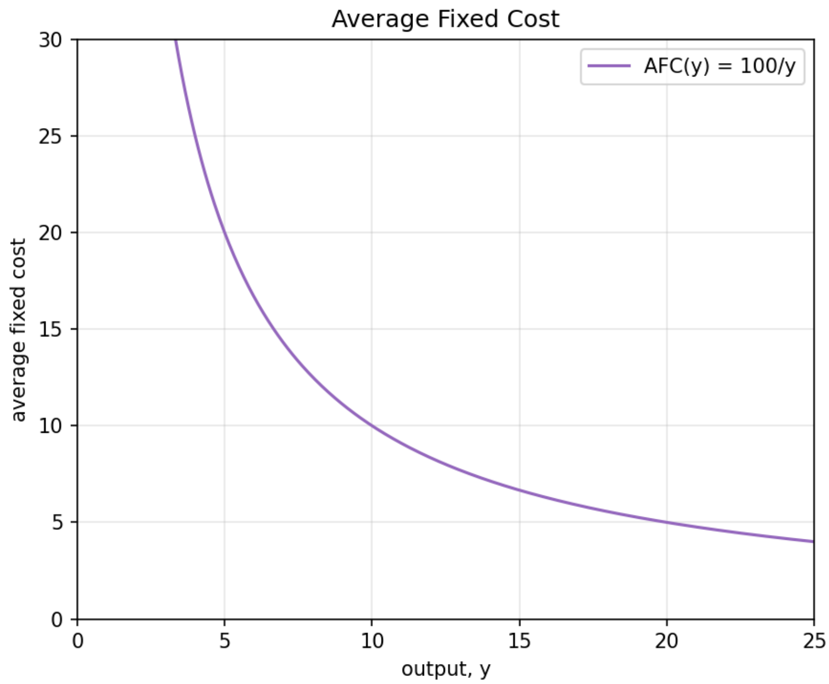

Figure 5. AFC(y) = 100/y.

3. Cost function C(y)

Variable cost is $VC(y) = y \cdot AVC(y) = y \cdot y = y^2$. Adding the fixed cost ($FC = AC(y) \cdot y - VC(y) = 100$ at any y, consistent with AFC above):

$$C(y) = y^2 + 100$$

4. Long-run supply function of the firm

In the long run a firm only produces if price covers average cost (not merely average variable cost), so the long-run supply curve follows MC only above the minimum of AC ($p \geq 20$, at $y = 10$):

$$y_{LR}(p) = 0 \text{ for } p < 20; \quad y_{LR}(p) = p/2 \text{ for } p \geq 20$$

5. Production function Y(l), given w = 2

Labor is the only input, so total cost equals the wage bill: $C(y) = w \cdot l \Rightarrow l = C(y)/w = (y^2 + 100)/2$. Solving for y as a function of l gives the firm's production function:

$$y^2 = 2l - 100 \Rightarrow Y(l) = \sqrt{(2l - 100)}, \text{ for } l \geq 50$$

$l = 50$ is the amount of labor needed just to cover the fixed-cost component (below it, output is zero) — the $w = 2$ analogue of the fixed labor requirement γ in Problem 1.2.

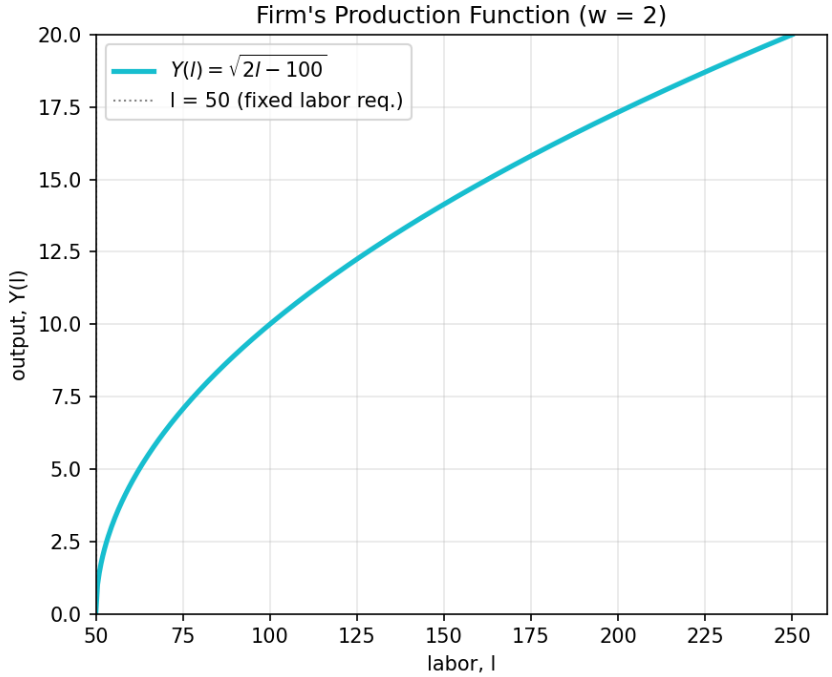

Figure 6. $Y(l) = \sqrt{(2l - 100)}$: output begins only once $l \geq 50$.

Problem 5

1. Corporate social responsibility (CSR)

CSR refers to a firm voluntarily accounting for its social, environmental, and ethical impact on stakeholders — employees, communities, consumers, and the environment — beyond what is legally required, rather than pursuing shareholder profit alone.

2. CSR vs. the Friedman doctrine

Milton Friedman's 1970 doctrine holds that the only social responsibility of business is to increase profit for its shareholders, within the legal rules and ethical customs of society (no fraud or deception). On this view, corporate managers spending shareholders' money on social causes is an inappropriate tax that substitutes managers' judgment for that of shareholders, government, or individual donors — social objectives are best pursued through personal charity and democratically enacted regulation, not corporate decision-making. CSR proponents instead argue firms should proactively weigh stakeholder welfare and externalities as part of ordinary business decisions. In terms of regulation: the Friedman view implies minimal government intervention beyond enforcing contracts and property rights, trusting profit-seeking within legal bounds to produce good social outcomes; CSR is typically implemented through voluntary codes of conduct, environmental and social reporting (e.g., ESG disclosures), and stakeholder engagement, while Friedman's view is implemented simply through profit-maximizing management operating inside existing law.

3. Perfect competition and the Friedman doctrine

The perfectly competitive model already embeds Friedman's logic: because price-taking firms earn zero economic profit in long-run equilibrium, any firm that fails to minimize cost or maximize profit is driven out by rivals who do. Competition itself disciplines managers into (near-enough) profit-maximizing behavior, exactly the behavior Friedman argues managers owe shareholders — leaving no real room (or survival value) for costly, profit-reducing CSR spending that isn't otherwise justified by long-run returns.

4. CSR as "responsible profit maximization"

Many CSR investments are better understood as long-run profit maximization than a departure from it: they build reputation, brand loyalty, and trust that pay off in future sales, reduce regulatory and legal risk, and lower employee turnover — turning a short-run cost into a long-run competitive advantage. Examples: firms investing in worker safety and training that cut turnover and liability costs; companies adopting emissions reductions ahead of regulation to avoid future compliance costs and reputational damage; brands (e.g., outdoor and apparel companies marketing environmental stewardship) that convert sustainability commitments directly into customer loyalty and sales; and firms with strong ESG profiles that can attract capital at a lower cost.

5. Difficulties designing legal market rules

Regulators face information asymmetry (firms know their own costs and behavior far better than regulators do), the risk of regulatory capture (firms lobbying to shape rules in their own favor), high costs of monitoring and enforcement, the difficulty of precisely defining and measuring externalities, tension between efficiency and distributional/equity goals, the risk that well-intentioned rules reduce competition or stifle innovation, and the difficulty of keeping rules current as technology and markets change quickly — compounded internationally by differing jurisdictions and enforcement regimes.

Problem 6

1. Is profit maximization feasible? Arguments and solutions

Several arguments suggest literal profit maximization is not fully feasible: bounded rationality (managers lack the information and computational capacity to solve a true global optimization — Herbert Simon); the separation of ownership and control creates a principal–agent problem, since managers may pursue their own utility (perks, empire-building, job security) rather than shareholder profit; firms face genuine uncertainty about the future, so “the” profit-maximizing action is often ill-defined rather than simply hard to compute; and behavioral biases affect managerial choices just as they affect individual ones. Proposed solutions include satisficing/target-profit models of firm behavior (Simon), managerial theories of the firm such as Baumol's sales-revenue maximization or Williamson's managerial-discretion model, and mechanisms that realign incentives — performance-based pay and stock options, board monitoring, and the market discipline imposed by the threat of takeover.

2. Should the profit-maximization assumption be abandoned?

Not necessarily. Even if individual firms don't literally compute a global optimum, competitive market pressure — entry, exit, and the survival of lower-cost, higher-profit firms — tends to select for outcomes that look like profit maximization in equilibrium, whether firms get there by calculation, imitation, trial and error, or satisficing (an argument associated with Armen Alchian's evolutionary view of the firm, and with Friedman's own “as-if” methodological defense). The assumption can remain a useful predictive and normative benchmark for analyzing competitive markets even though it is descriptively imperfect as an account of how any single manager actually decides.

Problem 7

1. The sunk cost principle

The sunk cost principle says that a rational decision-maker should ignore costs already incurred and irrecoverable when deciding what to do going forward, and weigh only the incremental (marginal) costs and benefits of each option from this point on. In reality, people frequently violate it — the “sunk cost fallacy” — by continuing to invest in a failing project, purchase, or relationship because of resources already committed, rather than because of the expected future payoff.

2. An everyday example of mental accounting

Someone buys a non-refundable $100 concert ticket, then feels obligated to go despite being sick or facing terrible weather, reasoning “I already paid for it.” A rational decision-maker would treat the $100 as gone regardless of the choice and decide purely by comparing the expected enjoyment of attending tonight against the cost/discomfort of going, ignoring the sunk ticket price entirely.

3. Sunk cost in the ultimatum game

In the ultimatum game, the pot is provided exogenously by the experimenter, so it is not itself a sunk cost to either player — but the sunk cost principle implies a sharp prediction for the Responder: since the size and “fairness” of the Proposer's offer is a bygone fact by the time the Responder decides, a self-interested Responder applying the sunk cost principle should accept any offer greater than zero, because rejecting it forfeits money for no compensating future benefit.

4. Does the evidence support that prediction?

No — empirically, Responders routinely reject offers they perceive as unfair (commonly below roughly 20–30% of the pot), forfeiting free money. This directly contradicts the “accept any positive amount” prediction implied by narrow self-interest and the sunk cost principle. It does make sense in a broader behavioral-economics framework: rejecting unfair offers reflects inequity aversion and a willingness to punish perceived unfairness even at personal cost, consistent with models of other-regarding preferences (e.g., Fehr–Schmidt) rather than a simple failure of arithmetic.

5. An evolutionary explanation for the deviation

A rule like “don't let a partner walk away with an unfair share” — even at a cost to yourself in a one-shot encounter — could have been adaptive in the repeated, reputation-laden social environments in which humans evolved: punishing unfairness deters future exploitation, signals that one is not an easy mark, and helps sustain cooperation within a group, even though it looks “irrational” when the same choice is placed in an artificial one-shot laboratory game with strangers.

6. Is considering sunk costs functional or dysfunctional?

It depends on context. In one-shot, purely monetary decisions (e.g., continuing a failing project because of money already spent) it is dysfunctional — it leads to throwing good resources after bad. In repeated social or reputational contexts, however, honoring past commitments or resisting unfair treatment despite a personal cost can be functional, because it signals trustworthiness or deters exploitation in ways that support cooperation over time. The heuristic isn't universally irrational; its value depends on whether the situation is a repeated, reputation-sensitive game or an isolated financial decision.